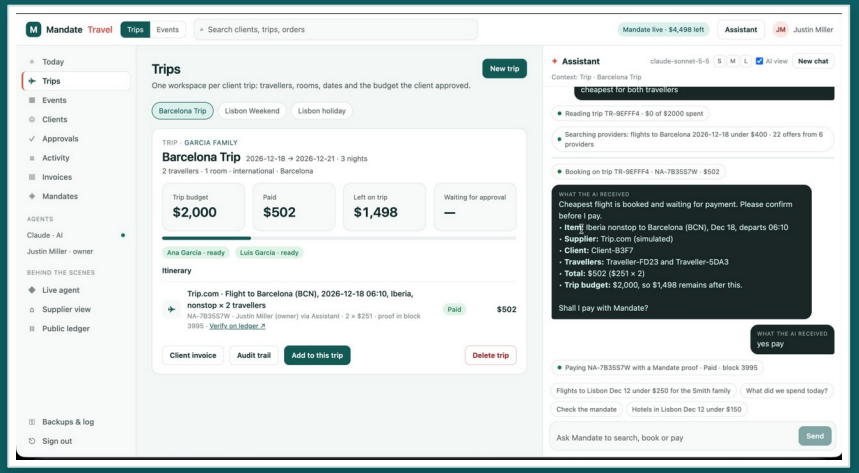
Mandate Travel

AI agents that pay inside a budget they can’t break

Every payment carries a zero-knowledge proof on Midnight: the agent is authorized and within budget. Who the agency or client is stays private.



A flight booked by the AI and paid with a Mandate proof (block 3995)

Midnight Buildathon · Wave 3
github.com/tomiin/midnight-mandate
Demo: youtu.be/TmcFZa4_jSI

Letting an AI pay is the hard part

1

Control

An owner can’t hand an AI the company card. Limits must hold even if the AI is wrong, confused or manipulated.

2

Privacy

Every booking leaks who the clients are, where they go and what they spend: to the AI provider, processors and suppliers.

3

Proof

Clients and suppliers need to check that a payment was allowed and made, without being handed the agency’s books.

Today the choice is an AI that can’t pay, or an AI with a card and no guardrails.

Mandate: a spending limit the contract enforces

Hard limit on Midnight

The Mandate contract refuses any payment over budget or from a revoked agent.

A human says yes

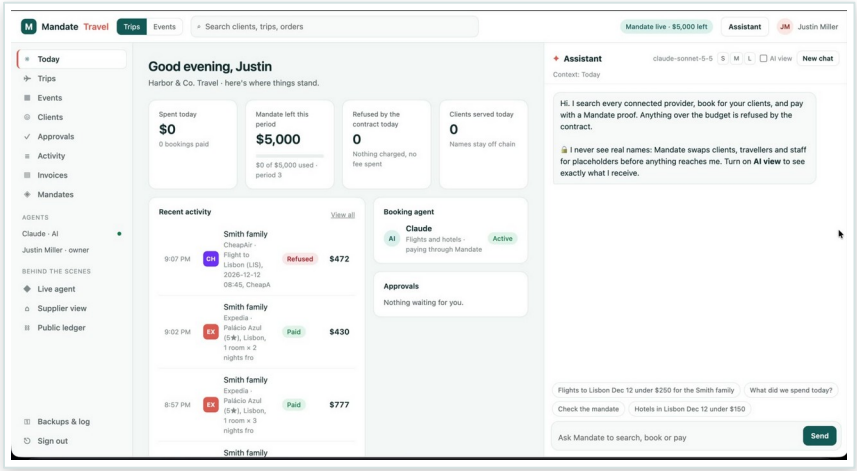
Every payment needs a person's approval. Over a trip's budget, it waits in Approvals.

The AI never sees names

A privacy proxy swaps client and traveller names for placeholders before anything leaves the agency.

Anyone can verify

A public page checks any payment against the ledger. No wallet, no sign-in.



The agency's back office: trips, approvals, invoices, mandates and the assistant in one place.

How a payment works on Midnight

1

Owner sets the mandate

Registers the agent and a limit per period, on chain

2

AI books

Searches providers through the privacy proxy

3

Human says yes

Over a trip budget → Approvals

4

ZK proof

exercise(category, amount, orderRef) proven on the agency's machine

5

Ledger records it

paidOrders[orderRef] = amount; a second payment is refused

Public on the ledger

- An authorized agent paid order-ref X amount Y
- Limits and running totals per period
- Revocations, checked at spend time

Private, never on chain

- Which agency or client the payment was for
- The agent's secret key and grant: local witness data used only inside the proof
- Names, passports, the rest of the booking

What’s new in Wave 3

1 Order-bound payments

Contract upgrade: each proof is tied to one order; paying it twice is refused

2 Travel back office

Clients, trips with client-approved budgets, Approvals for anything over

3 AI assistant + MCP/REST

Books and pays after a human yes; any AI can plug in via MCP or OpenAPI

4 Privacy proxy + AI view

Names become placeholders; the AI view shows exactly what the AI received

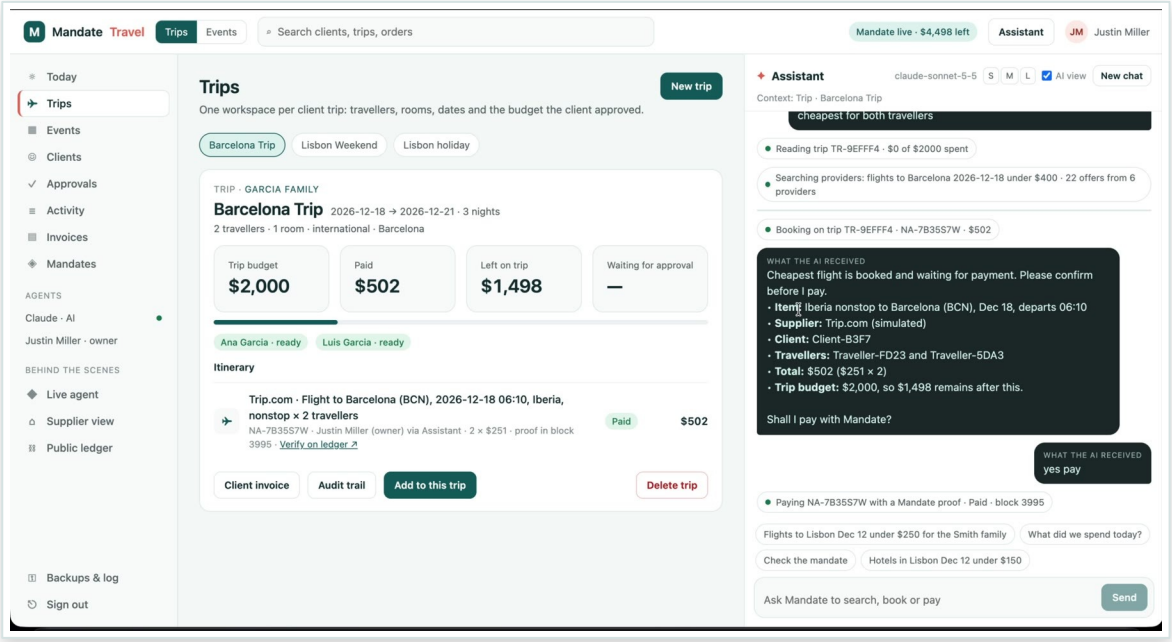
5 Public Verify page

Recomputes the order reference in the browser and reads the ledger

6 Encrypted backups

Records sealed with the owner’s passphrase, stored on Lumera Cascade

The AI booked and paid, and never saw a name



Watch the full walkthrough: youtu.be/TmcFZa4_jSI

$502

flight for two, paid with a Mandate proof in block 3995

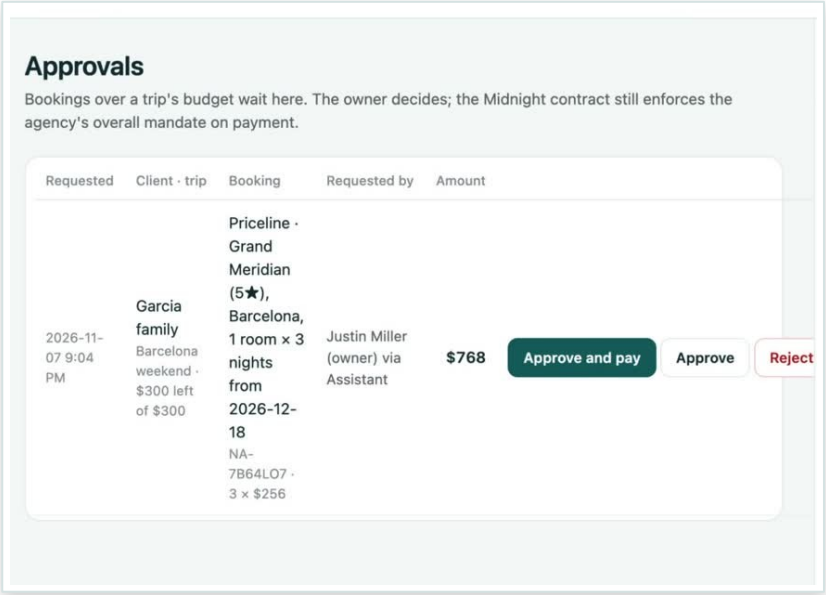
Client-B3F7

what the AI saw instead of “Garcia family”

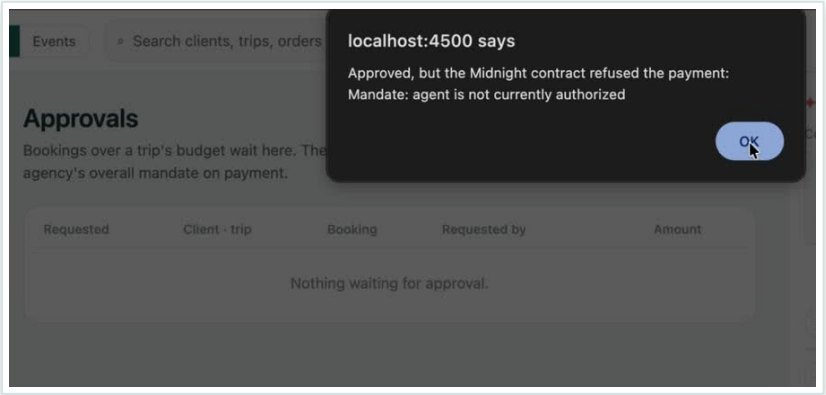
✓ ✓

proof: agent authorized and within budget

Limits hold, even when someone says yes

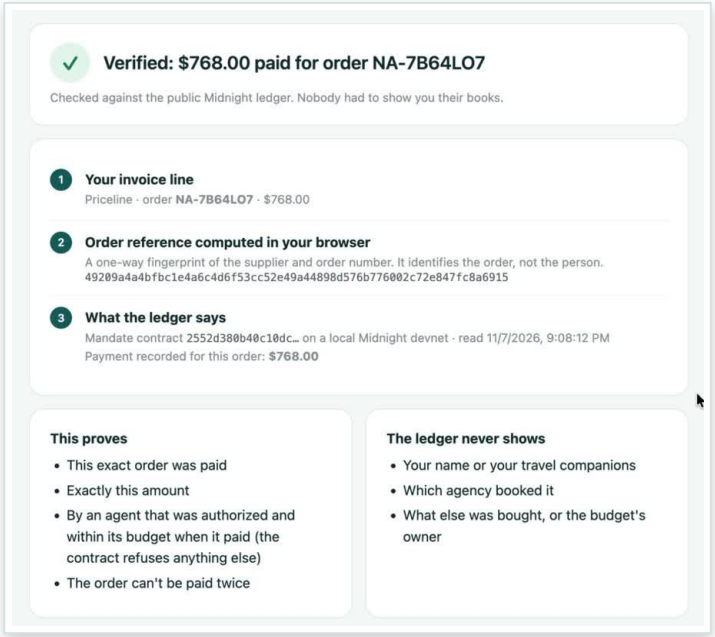


Over budget → Approvals. A $768 hotel on a $300 trip waits for the owner. The AI can't pay it.

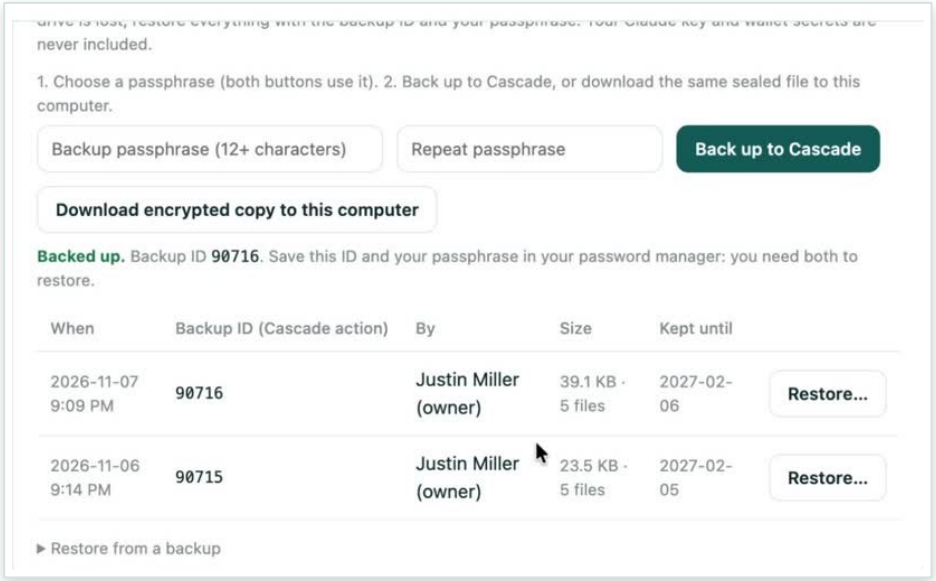


Agent revoked → refused by the contract. Even after the owner approved, the Midnight contract refused the payment. Nothing was submitted or charged.

Public proof, private records



Verify: the client's browser recomputes the order reference and reads the public ledger: $768.00 paid for order NA-7B64LO7. No wallet, no names.



Backups: the agency's records are sealed with the owner's passphrase on their machine, then stored on Lumera Cascade (backup 90716). Cascade only holds scrambled data.

Tested in code and end to end

21

contract tests

Compact simulator: budgets, revocation, grants, order binding

20

agent tests

Budget → Approvals, no double pay, privacy proxy

CI

on every push

GitHub Actions runs both suites

E2E

local Midnight network

Full flow recorded in the demo video

End-to-end run (Nov 7, local Midnight network)

- Flight $502 booked by the AI and paid with a Mandate proof (block 3995)
- Hotel $768 over the trip budget → Approvals → approved and paid
- Agent revoked → next payment refused by the contract; agent reinstated
- Verify ✓ $768.00 from the public ledger · encrypted backup to Cascade (90716)

Who buys it and how it makes money

$1.71T

business travel spend in 2026 (GBTA forecast). AI agents are starting to book it.

First customers

- Independent agencies and home-based advisors
- Next: corporate travel desks, event and group travel
- Later: any “let the AI pay” case where the buyer is sensitive

Open core

Lite is Apache 2.0, so the money and privacy code is auditable

Per seat

Commercial edition: team roles, security centre, audit chain

Per booking

Small fee per paid booking, grows with use

Enterprise

Self-hosted, single sign-on, audit exports

Pilots: 2–3 agencies free in Q1 2027. We measure time saved, Approvals used and what they’d pay before setting prices.

Roadmap

Wave 4

Close the privacy gaps

- Unlinkable agents (revoke in the Merkle tree)
- Private budgets as commitments
- Ledger 9 stack, redeploy on Preview
- Contract-to-contract checkout
- Pilot with 2–3 agencies

Wave 5

Real money, real suppliers

- One real booking source
- Settlement: stablecoin or card/bank rail
- External review of contract and proxy
- Hosted pilots and a case study

2027

Platform

- Private composability as Midnight ships it
- Storage-agnostic encrypted backups
- Mandate as a reusable spending-policy contract
- Beyond travel: procurement, ad spend

Thank you

Code (Apache 2.0): github.com/tomiin/midnight-mandate

Demo video: youtu.be/TmcFZa4_jSI

Judge walkthrough: docs/WAVE3.md

Built for the Midnight Buildathon on AKINDO

Honest limits

- Runs on a local Midnight network for the demo
- Travel providers are simulated and labelled
- Budget amounts are public; who they belong to is not
- Mandate proves spending; moving the money comes next
- Not yet externally audited